

SAMUEL WANG

Pleasanton, CA | (925)-368-0001 | samuel.t.wang@vanderbilt.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Vanderbilt University

Nashville, TN

Bachelor of Engineering in Electrical and Computer Engineering, Minor in Computer Science

May 2027

- **Relevant Coursework:** VLSI Design, Computer Architecture, Microcontrollers, IC Technology & Fabrication
- **GPA:** 3.97, Dean's List (all semesters)

EXPERIENCE

Marvell

May 2026 – Present

Analog/Mixed-Signal CAD Intern

Santa Clara, CA

- Engineered AI-assisted agentic automation tools within Cadence Virtuoso using Python and SKILL to streamline complex analog/mixed-signal simulation workflows
- Optimized simulation reliability and speed by developing tools for testbench automation, corner configuration, and pre- and post-simulation diagnostics, catching >80% of errors
- Served as Intern Peer Leader, organizing and facilitating events and engagement activities for 100+ interns

Vanderbilt University School of Engineering

Aug 2023 – Present

Undergraduate Research Assistant

Nashville, TN

- Lead author on research paper evaluating SRAM- and D flip-flop-based Physical Unclonable Functions (PUFs) in 3- and 5-nm FinFET technologies, performing data processing and statistical analysis for hardware security applications (poster presentation at IEEE IRPS conference in March 2026)
- Characterizing BTI-induced degradation in 3-nm ring oscillator circuits by measuring and analyzing long-term ring oscillator frequency drift under sustained bias and temperature stress (ongoing)

General Motors

May 2025 – July 2025

Controls Engineer Intern

Lansing, MI

- Developed a Python-based diagnostic and verification GUI tool for real-time telemetry data visualization, reducing debugging and verification time from hours to minutes and accelerating anomaly detection during live runs
- Built a CAN-based automated hardware test and validation setup for manufacturing equipment, improving test throughput by 65% across 3 production lines

PROJECTS

RISC-V RV32I Processor | *SystemVerilog*

May 2025 – Current

- Designed a synthesizable RV32I in-order RISC-V processor, verified using testbenches in Icarus Verilog to validate instruction fetch, decode, execute, memory, and writeback datapaths
- Expanding the processor into a 5-stage pipelined architecture featuring hardware hazard detection, data forwarding logic, and microarchitectural optimizations

FPGA-Based Minesweeper Game Engine | *VHDL, FPGA Design Final Project*

Mar 2026 – May 2026

- Co-designed a hardware-accelerated game engine on an FPGA using VHDL RTL, implementing an FSM controller, BRAM grid storage, parallel combinational adjacency logic, and an LFSR for random mine generation

8x8 Radix-4 Booth Multiplier | *Cadence Virtuoso, VLSI Design Final Project*

Nov 2025 – Dec 2025

- Designed and verified an 8-bit Radix-4 Booth multiplier from schematic to full custom layout, reducing partial products by 50% and utilizing a Carry-Lookahead Adder (CLA) for high-speed summation
- Achieved DRC/LVS cleanliness and performed parasitic extraction (PEX) to run post-layout timing simulations and validate the design

TECHNICAL SKILLS

Programming: Verilog, SystemVerilog, VHDL, SKILL, Python, C/C++, MATLAB

EDA & Development Tools: Cadence Virtuoso, GTKWave, LTspice, Git/GitHub

Hardware: FPGAs, STM32, Raspberry Pi, Oscilloscopes, PCB Design